

# Ravindu Lakshan

Department of Computer Engineering, University of Peradeniya, Sri Lanka

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## PROFILE

I am a motivated **4th-year Computer Engineering undergraduate** with a broad and applied technical skill set. My core academic interests are rooted in **Computer Architecture**, **Embedded Systems**, and futuristic fields like **Neuromorphic Computing**. I possess hands-on proficiency in advanced software domains, including **Machine Learning**, **Computer Vision**, and **Web/Software Development**. I am passionate about leveraging this diverse knowledge base to engineer innovative solutions, and I bring strong and collaborative **teamwork** capabilities to challenging projects.

## EDUCATION

### University of Peradeniya

*Undergraduate in B.Sc. Engineering (Hons.) Computer Engineering*

*Current GPA: 3.61/4.0*

*Peradeniya, Sri Lanka*

*Nov. 2022 – Present*

### B/Orubediwewa Central Collage

*G.C.E. Advanced Level Examination*

*Z-score: 2.0596 — National Rank: 831/29737 — District Rank: 12/1947*

*Mahiyanganaya, Sri Lanka*

*Jun. 2018 – Oct. 2020*

## PROJECTS

### On-Chip Offline Learning for Neuromorphic Accelerator | *SNN, Verilog, RISC-V, FPGA* Aug. 2025 – Present

- Investigating an energy-efficient neuromorphic accelerator pipeline for spiking neural networks (SNNs) with event-driven computation and local synaptic updates.
- Designing an on-chip offline learning workflow that updates synaptic parameters using locally available spike timing/activity signals, reducing dependency on host-side training.
- Prototyping a configurable architecture (neurons, synapse memory, and learning control) to explore trade-offs among memory bandwidth, update precision, and energy per inference/training step.

### Safe Plus - Smart Safety Helmet | *Arduino/ESP32, MQTT, AWS IoT, MERN* | Nov. 2024 – Jul. 2025

- Designed an advanced smart safety helmet integrating real-time monitoring, impact detection, and emergency alert systems for industrial workplaces.
- Enhanced worker safety by enabling immediate hazard detection (toxic gas, impacts, falls) and automated supervisor connectivity via the dashboard.
- My Contribution:** Handled complete hardware development, including sensor integration into the helmet, and wrote firmware for all real-time monitoring features.

### Denture Design Studio | *React.js, Next.js, Express.js, MongoDB* | Aug. 2024 – Nov. 2024

- Developed a graphical platform for real-time denture design practice for dental students, providing a safe learning environment.
- Implemented features for self-evaluation, allowing students to compare their designs with assessor-provided model outputs.
- My Contribution:** Implemented student activity tracking, developed the Assessor page frontend, and built the backend logic using Node.js/Express.js.

### CO502 RV32IM Pipeline | *Verilog HDL, GTKWave, Synopsys EDA tool* | Nov. 2024 – Mar. 2025

- Implemented a RISC-V (RV32IM) processor pipeline in Verilog, including stages for instruction fetch, decode, execute, memory access, and write-back.
- Developed a comprehensive testbench to verify processor functionality and ensure adherence to timing constraints.

### Real-Time Shadow Removal | *Python, OpenCV, TensorFlow/PyTorch* | Nov. 2024 – Mar. 2025

- Developed an image processing system for real-time shadow removal to improve visual quality and enhance object detection in real-world applications.
- Employed techniques for shadow region detection (color gradients, texture) and image enhancement to fill shadow areas.

**NeuroDrive(Vehicle Detection System)** | *Python, YOLOv8, Streamlit, OpenCV* |  Oct. 2025 – Jan. 2026

- Built an end-to-end computer vision application for vehicle detection and classification, providing both single-image inference and batch folder processing via a Streamlit UI.
- trained and evaluated a custom YOLOv8 model across 8 vehicle classes (car, bus, truck, motorcycle, bicycle, three-wheeler, tractor, van), achieving best validation mAP@0.5 of 80.6% and mAP@0.5:0.95 of 63.8%.
- Implemented automated outputs including annotated images, per-class distribution visualizations, and downloadable CSV reports to support analysis and reporting.

## EXPERIENCE

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### Casual Instructor

Feb. 2024 – Jul. 2025

*Department of Computer Engineering, University of Peradeniya*

*Peradeniya, Sri Lanka*

- Supervised weekly 3-hour lab sessions for the **GP106** and **CO224** courses, effectively assisting students in applying knowledge of **Python**, **assembly** and **Verilog** languages.
- Helped students understand complex course material and strengthened their foundational programming skills.

## VOLUNTEERING & LEADERSHIP

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### Volunteer(Nenathmbara program)

*Project Nenathambara, University of Peradeniya*

*Peradeniya, Sri Lanka*

- Coordinated with the organizing committee for **Project Nenathambara**, an Arduino-based competition hosted by the Department of Computer Engineering for school students.
- Contributed to the planning and execution of competition logistics, judging, and event volunteering.
- Contributed to the organization and facilitation of preparatory **workshops on Arduino and robotics** for underprivileged students.

### Volunteer (Arunalla Program)

*Faculty of Engineering, University of Peradeniya*

- Volunteered in the Arunalla Program, a significant **Maths/Science seminar series** for O/L students in **rural areas**.

### Vice President / Organizer

*Hackers Club / Web Development & ML Series*

*Peradeniya, Sri Lanka*

- Served as **Vice President** of the Hackers Club, managing executive functions and coordinating technical skill development events.
- **Organized and facilitated** training series in **Web/Mobile Development** and **Machine Learning (ML)**.
- Served as a **Resource Person** for the Web Development Series, teaching topics like **HTML/CSS, JavaScript, and React.js**.

## TECHNICAL SKILLS

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**Languages:** Java, Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS, Verilog HDL

**Frameworks:** React, Node.js, Express.js, Streamlit, Flask, JUnit

**Developer Tools:** Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Synopsys EDA,

**Libraries:** pandas, NumPy, Matplotlib